MICHELE MOTTA

Italian Institute of Technology (IIT), Center for Life Nano Science, Rome, Italy

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LANGUAGES

Fluent in Italian (native) and English

WORK EXPERIENCE

(June -)	**PhD Student**
2026 -	Project: ‘**Coherent long-range neuronal oscillations at voice pitch frequencies: a neurophysiological mechanism of fundamental communication in phonatory control**’
	BrainLab – Institute of Neurosciences, Universidad de Barcelona
(October - June)	**Research Assistant**
	Project: ‘**Electrospinography analyses in time and frequency domains**’
2025 - 2026	Neuroscience and Behavior Laboratory, IIT, Rome, Italy (PI: Giandomenico Iannetti)
(May - September)	**Research Assistant**
	Project: ‘**Effective connectivity of the mirror neuron system**’
2025	Laboratory for Neuro- and Psychophysiology, KU Leuven, Belgium (Supervisor : Sara Riva; PI: Peter Janssen)
	Acquired Skills: • Electrical microstimulation during fMRI with acute and chronic electrodes in both awake and sedated monkeys • Reversible inactivation with muscimol during fMRI • Preprocessing and statistical modeling and analysis of block-design fMRI data with SPM • Assistant during single-cell recordings in anterior intraparietal cortex • Monkeys’ training for grasping • Assistant in chambers’ cleaning for single-cell recording sessions

EDUCATION

2023 - 2025	Master’s degree in **Cognitive Neuroscience**
	Experimental thesis: ‘**Behavioral Investigation of Heuristic Behavior in a rule-based foraging task in NHPs**’
	‘La Sapienza’ University of Rome, Italy
2020 - 2023	Bachelor’s degree in **Psychology**
	Experimental thesis: ‘**Multisensory integration in patients affected by obsessive-compulsive disorder**’
	University of Parma, University of Modena and Reggio Emilia, Italy

LAB EXPERIENCE

2023 - 2025	Reseach Trainee under the supervision of Prof. Ferraina and Dr. Kirchherr

	Project: '**Behavioral Investigation of Heuristic Behavior in a rule-based foraging task in NHPs**'
	Department of Physiology and Pharmacology, 'La Sapienza' University of Rome (PI: Stefano Ferraina)
	Acquired Skills: • Chronic recording with Utah Arrays in pre-frontal and motor areas (DLPFC, PMd) • Macaque handling (positioning on the primate chair, caging after the recording session, cage interactions, wound cleaning) • Development and debugging of Matlab codes for behavioral analysis (saccades, pupil, touch reaction time) together with analysis of the strategy implemented by the animal and the creation of stimuli in MonkeyLogic • Data and statistical analysis in Matlab and Python
2024 - 2025	Research Trainee under the supervision of Prof. Giandomenico Iannetti
	Project: '**Revamping Clynes: Pitch change inhibition in R – M response**'
	Neuroscience and Behavior Lab, Italian Institute of Technology (PI: Prof. Gaindomenico Iannetti)
	Acquired Skills: • Design of an EEG study on the influence of pitch and volumes changes in auditory-evoked vertex potential • In-depth understanding of advanced EEG theory and first-hand experience in EEG recordings • Development of Python scripts for the analysis of EEG data • Implementation of the setup for neural signal recording (creation of stimuli in Python, modulation of sound frequency, assessment through in-ear headphones) • Data collection with human subjects.
2024 - 2025	Research Trainee under the supervision of Prof. Luca Simione
	Project: '**Effects of Mindfulness Meditation on the bodily self**'
	Institute of Cognitive Sciences and Technologies – CNR (PI: Luca Simione & Antonino Raffone)
	Acquired Skills: • Behavioral data analysis (reaction time) • Scoring of psychological questionnaires (MAIA, PS, PHMLS) • Data collection with human subjects • Extensive literature reading on the topic of peripersonal space
2022 - 2023	Research Trainee under the supervision of Prof. Martina Ardizzi
	Project: **'Multisensory Integration in patients affected by obsessive-compulsive disorder (OCD)'**
	Social Cognitive Neuroscience Lab, University of Parma (Supervisor: Martina Ardizzi; PI: Prof. Vittorio Gallese)
	Acquired Skills: • Introduction of patients with OCD to the experiment • Implementation of the setup for behavioral recordings (running the experiment and recordings to assess patients' reaction time) • Positioning of electrodes for both tactile and visual stimuli stimulation on participants' hands, setting of electrical stimulation according to participants' pain threshold

	• Scoring of psychological questionnaires (Y-BOCS, CTQ, TCI, BABS) • Data collection with patients affected by OCD • Extensive literature reading on the topic of multisensory integration and OCD

PROGRAMMING AND SOFTWARES

Matlab; Python; HTML; MonkeyLogic; SPM; Letswave; MRIcron; 3D Slicer;

METHODS

Invasive neural recordings with Utah Arrays
Single cell recording with platinum-iridium electrodes
Stimulation during functional magnetic resonance imaging (fMRI) with microelectrodes
Inactivation with muscimol during fMRI
Electroencephalogram (EEG)
Electrophysiological signal processing (EEG, LFPs)
Pre-processing and processing of fMRI data

CERTIFICATES

(Italian) animal experimentation certificate (non-human primates, mice)
'La Sapienza' University of Rome, Italy
Felasa B – Animal experimentation certificate (non-human primates, mice)
KU Leuven, Belgium

BEYOND SCIENCE

Writing, Running, Reading, Cycling, Windsurfing, Cinema